

COMPUTER ENGINEERING / AI SYSTEMS / PRODUCT SOFTWARE

Computer engineering, built systems.

Mohammad J. Bakr builds practical software where automation, interfaces, and safety have to work together.

SAUDI ARABIA

OPEN TO ENGINEERING ROLES

A portfolio of local-first systems, Arabic-first product interfaces, automation pipelines, and motion experiments.

PYTHON

TYPESCRIPT

REACT

SQLITE

AUTOMATION

ARABIC UI

mohammadbakrwork@gmail.com

GITHUB.COM/M7MDBKR

THE THROUGHLINE

Systems that stay understandable.

The projects vary in surface area — an operating system, a POS, a news bot, a route planner, a video pipeline — but the engineering instinct stays consistent.

- 01

MAKE STATE VISIBLE

Dashboards, timelines, logs, and clear empty states.
- 02

KEEP ACTIONS EXPLICIT

Approval gates before email, publishing, or deployment.
- 03

DESIGN FOR THE REAL USER

Arabic-first flows, local constraints, and practical defaults.
- 04

LEAVE A TRAIL

Tests, manifests, source records, and honest limitations.

BUILD MAP

FROM IDEA

brief / problem / constraint

TO SYSTEM

data / policy / interface

TO PROOF

tests / smoke check / review

No cloud dependency by default.
No silent external side effects.
No claims without a visible trail.

01 / FEATURED CASE

AgenticOS

A local-first personal agent operating system.

Four specialist workspaces, a human-in-the-loop approval layer, Telegram and n8n hooks, local context, schedules, page-agent interactions, and a SHA-256 audit chain.

THE ENGINEERING STORY

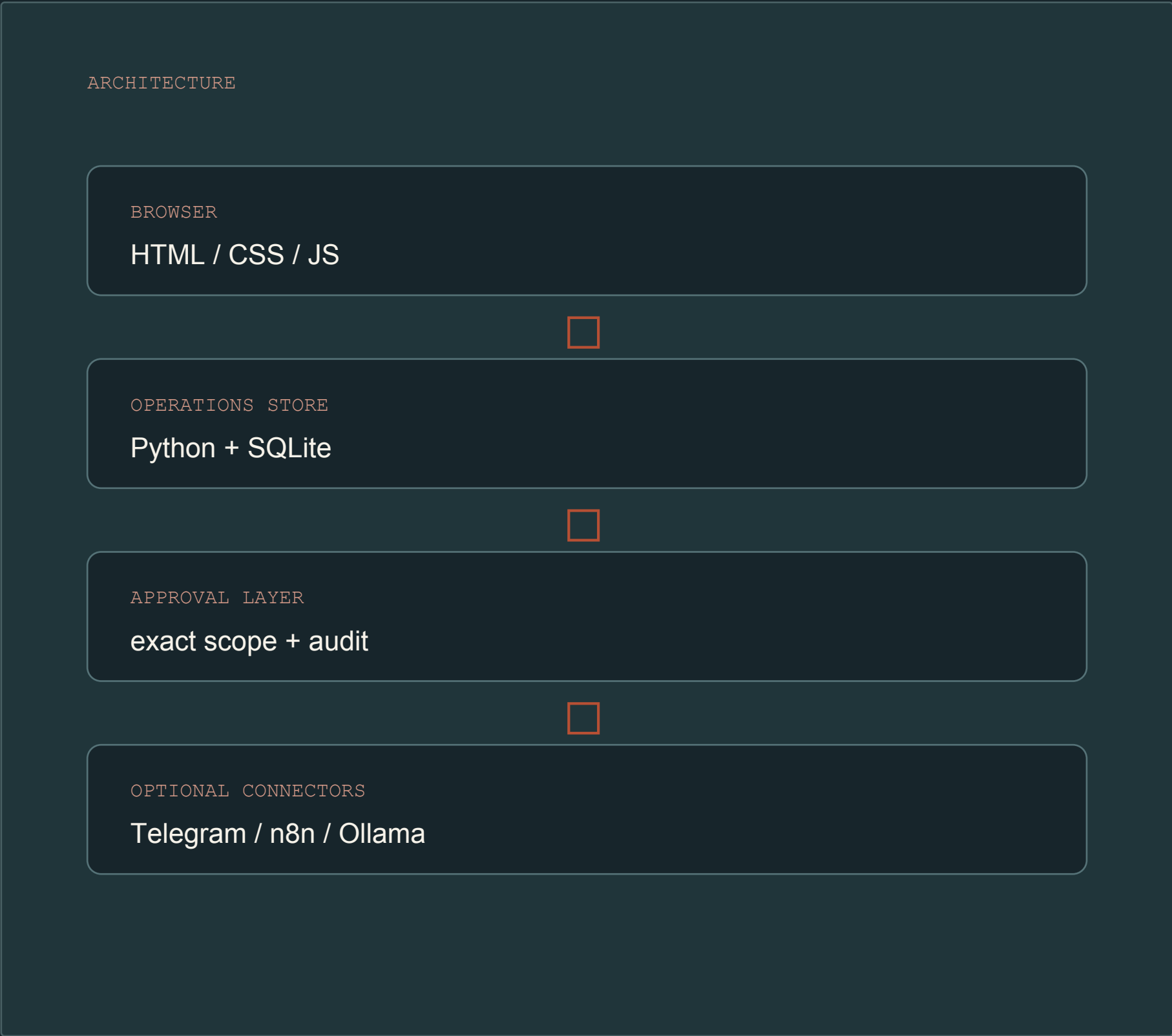
- State is stored locally in SQLite.
- External actions become reviewable cards.
- Providers are replaceable instead of permanent.
- Runtime data stays outside the public snapshot.

16 TESTS PASS

PYTHON + JS SYNTAX

LOOPBACK SMOKE TEST

github.com/M7mdbkr/agenticos



PUBLIC SNAPSHOT · RUNTIME DATA EXCLUDED

02 / PRODUCT CASE

Rizq POS

Arabic-first point of sale for cafés and restaurants.



THE PRODUCT STORY

A real operational surface,
not just a dashboard mockup.

Separate cashier and owner flows.

Browser-local SQLite/WASM storage.

Receipts, CSV export, inventory, VAT.

RTL layout and Arabic typography.

HONEST READINESS NOTE

Build passes; security hardening is required before real customer data.

REACT

TYPESCRIPT

VITE

SQL.JS

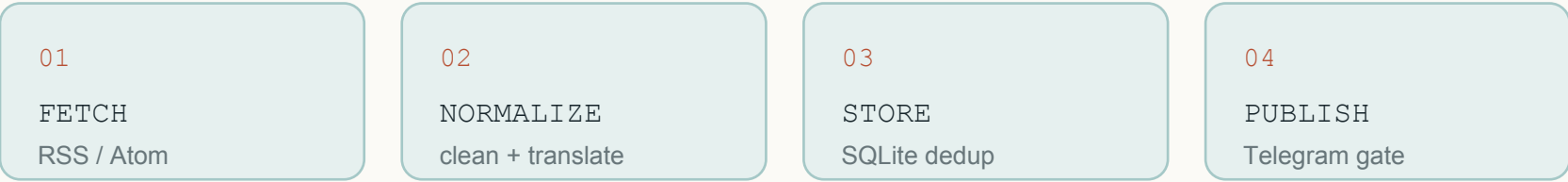
github.com/M7mdbkr/rizq-pos

Pipelines that move from signal to useful output.

ARABIC NEWS BOT

RSS → Arabic → Telegram

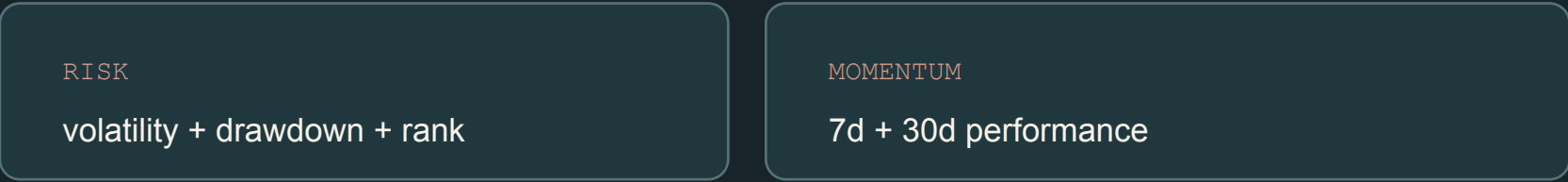
Cars and technology feeds become deduplicated stories, Arabic headlines, owner-only commands, and weekly reports plus ready-to-edit video scripts.



CRYPTO SIGNAL ANALYZER

Data, with the caveat attached.

CoinGecko market data becomes explainable indicators: annualized volatility, drawdown, market-cap rank, and recent momentum.



The point is not “AI magic.” It is a chain a teammate can inspect, run, and improve.

Make the next action obvious.

AOS CORE

OPERATIONS EXPLORER

Interactive architecture map for agents, task routing, exact-scope approvals, audit events, and a loopback-only API.

PYTHON

SQLITE

POLICY

ROUTE PLANNING

MALHAM LINE

Editorial route-planning interface with stops, timings, mode changes, bilingual context, and light/dark themes.

HTML

SVG

ARABIC UI

WORKFLOW UX

JOB SEARCH DASHBOARD

Bilingual job-search workspace concept: new matches, drafts, awaiting replies, applications, and assistant flow.

HTML

BILINGUAL

PRODUCT UX

Design is not decoration here. It is the compression layer between system state and human decision.

DESIGN-LED · LOCAL-FIRST · READABLE

05 / MOTION SYSTEMS

Motion as a system, not a pile of clips.

Remotion work across car edits, Arabic tech news, commercial creative, and children’s storytelling. The code separates timing, composition, and assets.

01	CAR MEET	clip timing · camera movement · audio	
02	TECH NEWS	captions · voice layers · vertical framing	
03	WISAA	product story · Arabic type · campaign rhythm	
04	LITTLE LANTERNS	storyboard · hooks · QA manifest	
05	WHEELS ON THE BUS	FFmpeg assembly · deterministic render	

PUBLIC SNAPSHOT RULE · SOURCE CODE is shareable; generated media and imported assets stay out until rights are clear.

A clean public snapshot is part of the work.

Evidence captured before publication

PROJECT	PROOF	PUBLIC POSTURE
AgenticOS	16 unit tests · syntax · smoke test	ready / runtime excluded
Operations Explorer	23 core tests · API policy checks	ready / local core
Rizq POS	npm run build passes	hardening before real use
Smart Black Box	C++ / ESP32 source on GitHub	private / access on request
Remotion projects	lint + TypeScript checks pass	source-only / assets excluded

Interfaces + analysis

static parse + source audit

docs + prototypes

LEFT OUT OF THE PUBLIC SET

credentials · runtime databases · browser profiles
private vault notes · generated media · third-party clones

WHAT A COMPANY CAN SEE

problem framing · architecture · code quality
tests · constraints · honest readiness notes

LET'S BUILD THE

next useful system.

Computer engineering student at Taif University, graduating in 2026. Interested in AI systems, automation, product software, embedded systems, and cloud/networking foundations.

CONTACT

Mohammad J. Bakr

mohammadbakrwork@gmail.com

github.com/M7mdbkr

Saudi Arabia · Arabic / English

OPEN TO ENGINEERING ROLES

AI SYSTEMS

AUTOMATION

PRODUCT SOFTWARE

INTERFACE DESIGN

Thank you for looking.